

Project 1: Exploratory Data Analysis of Historical Stock Market Data

Question: what does roughly half a century of daily US stock data look like, how clean is it, and which unusual values are real market events versus data problems?

The price file is large (20,973,889 rows, 1970-01-02 to 2018-08-24) and structurally clean: no missing values and no duplicates. The problems are hidden in the values. Prices are extremely right-skewed (mean close \$76.11 against a median of \$15.45), and a small group of tickers holds impossible or extreme numbers. The worst is AAN in early 1987, where **adj_close** reaches 1.89×10^{19} while **close** is about \$1.30. The stocks file also contained one accidental header row, which was removed. After investigating each case, 4,060 rows (0.02%) were excluded from price analysis. With the median instead of the mean, the cleaned data clearly shows the 1973–74, 1987, 2000–02 and 2007–09 market declines.

This is an exploratory analysis of historical data. Nothing in this report is financial advice.

1. Data and Structure

Two files were analysed. They are linked through the **ticker** column, a short code that uniquely identifies each security (for example KO for Coca-Cola). One security in **historical_stocks** matches many daily rows in **historical_stock_prices**, so the relationship is one-to-many.

Item	historical_stocks.csv	historical_stock_prices.csv
What it represents	Reference list of securities: one row per ticker	Daily trading record: one row per ticker per day
Rows	6,460 as loaded (6,459 after removing the header row)	20,973,889
Columns	5	8
Numerical columns	None	open, high, low, close, adj_close, volume (float64)
Categorical columns	ticker, exchange, name, sector, industry	ticker
Date column	None	date (text on load, converted to datetime)
Columns with missing values	sector, industry	None
Memory	252.5 KB	1.3 GB

2. Data Quality

2.1 Missing values

The price file has zero missing values in all eight columns. The stocks file is missing in **sector** and **industry** for the same 1,440 securities (22% of the list).

Dataset	Column	Missing	Decision
historical_stocks	sector	1,440	Kept. The security itself is valid.
historical_stocks	industry	1,440	Kept. Same rows as sector.

Dataset	Column	Missing	Decision
historical_stock_prices	all columns	0	No action needed.

Are missing values necessarily errors? No. Missing sector often means the source never classified that security. Many of the unmatched tickers (ASA, ADX, GAB, JHI, MXF) are closed-end funds and trusts, which do not belong to an industry sector the way an operating company does.

Should every missing value be replaced? No. Filling sector with a guess such as the most common value (FINANCE) would invent information. Deleting the rows would throw away 1,440 real securities and their price history. Leaving them as missing, or labelling them "Unknown" for charts, keeps the data honest.

2.2 Duplicates

Both checks returned zero: there are no fully duplicated rows and no repeated **ticker + date** combinations.

Why check ticker + date separately? A full-row check only catches rows that are identical in every column. Two rows for AAPL on the same day with slightly different prices would pass that check, yet one stock can only have one record per trading day. Ticker + date is the business key, so it is the stricter and more meaningful test.

2.3 Header-like records

A search for ticker values containing the word "ticker" returned nothing, and no ticker is longer than six characters. However, the first sector chart showed a category literally named **SECTOR**. Filtering on it found exactly one record:

Index	ticker	exchange	name	sector	industry
74	Symbol	NYSE	NAME	SECTOR	INDUSTRY

Every value is a column label, not information about a company, so this is an accidental header row (probably from a second file that was appended with its own header). It was removed. The stocks file now has **6,459 rows and 12 real sectors**, instead of the 13 reported by `.describe()`. The 1,440 securities with a blank sector are genuine and were kept.

Obvious error vs. missing value? An accidental header is wrong information: its values describe the file, not a security, so keeping it corrupts counts. A missing value is absent information about a real record. The first is removed; the second is kept and documented.

3. Date Conversion (Part 6)

The **date** column was converted with `pd.to_datetime()` (no values failed to convert) and set as a sorted index. The **earliest date is 1970-01-02** and the **latest is 2018-08-24**. The data therefore ends in August 2018, not in 2020 as the dataset title suggests.

Stored as text, "2018-08-24" is just a string: it sorts alphabetically and cannot be grouped by month. As a datetime it supports resampling to monthly values, extracting the year to build the decade column, filtering by date range, and plotting on a real time axis. All the time-series work in sections 8 and 9 depends on this step.

Decade	Rows	Tickers
1970s	164,499	93

Decade	Rows	Tickers
1980s	1,470,571	901
1990s	3,427,832	2,117
2000s	6,572,053	3,429
2010s (to Aug 2018)	9,338,934	5,685

The number of tickers grows 61× from the 1970s to the 2010s. Each decade describes a different set of companies, which matters for every comparison later in the report.

4. Categorical Variables

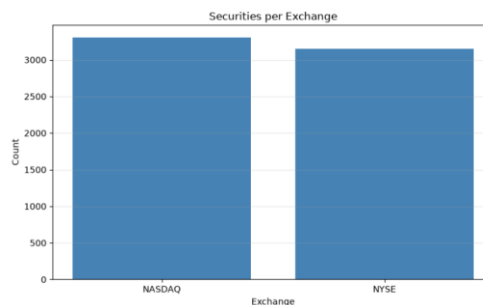


Figure 1. Number of securities per exchange.

Two exchanges are represented. **NASDAQ** lists 3,308 securities and **NYSE** 3,151 (after the header row, which was labelled NYSE, was removed), so NASDAQ has slightly more (51%).

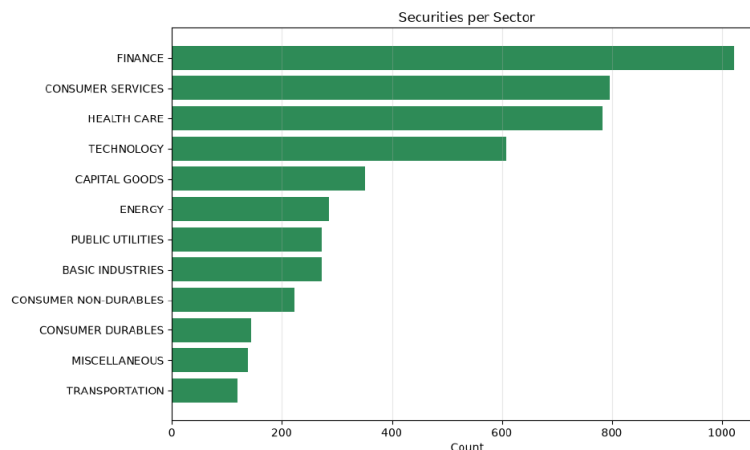


Figure 2. Number of securities per sector, after removing the header row.

FINANCE is the largest sector with 1,022 securities, followed by Consumer Services and Health Care (close to 800 each) and Technology (about 600). The most common industry is Major Pharmaceuticals (419). 1,440 securities have no sector. Likely reasons are that they are funds, trusts or ETFs rather than operating companies, that they were listed recently and never classified, or that the source simply had no record for them.

When the two files are joined on ticker (left join), the row count stays at 20,973,889, confirming no row inflation. 2,549,449 price rows (12.2%) end up with no sector.

5. Numerical Variables

Column	Mean	Std	Min	25%	Median	75%	Max
open	76.06	2,849.64	0.0004	7.50	15.45	29.72	2,034,000
high	78.04	2,997.94	0.0004	7.63	15.66	30.10	2,070,000
low	74.22	2,746.06	0.0001	7.36	15.24	29.28	1,440,000
close	76.11	2,870.16	0.0002	7.50	15.45	29.72	1,779,750
volume	1,227,043	13,166,860	1	22,100	126,000	607,400	4,483,504,000
adj_close	1.48e14	4.57e16	2.3e-09	4.62	11.38	24.72	1.89e19

Count is 20,973,889 for every column.

Median vs. mean close: the median close is **\$15.45** and the mean is **\$76.11**, about 4.9 times higher. Half of all trading days closed below \$15.45.

Largest standard deviation: among the five required columns, **volume** (13.2 million shares). **adj_close** is even larger (4.57×10^{16}), but that value is driven by broken records (section 7).

Large maximums: all four price columns reach over \$1.4 million, volume reaches 4.48 billion shares, and **adj_close** reaches 1.89×10^{19} . The 75th percentile of close is only \$29.72, so these maximums are tens of thousands of times the typical value.

What the gap tells us: when the mean sits far above the median, a few very large values are pulling it up. The distribution is strongly right-skewed, and the median is the better description of a "typical" stock.

6. Highest Closing Prices

Ticker	Date	Open	High	Low	Close	Volume
SCON	2000-02-28	1,395,000	1,797,750	1,368,000	1,779,750	400
SCON	2000-02-29	2,034,000	2,070,000	1,440,000	1,595,250	400
SCON	2000-03-13	900,000	1,404,000	861,750	1,363,500	300
CODA	2001-02-05	1,354,500	1,736,700	1,354,500	1,354,500	34
TVIX	2013-02-26	1,375,000	1,542,500	1,300,000	1,347,500	100
SCON	2000-03-02	1,120,500	1,368,000	1,098,281	1,261,125	200
TOPS	2016-08-01	783,000	1,276,200	783,000	1,189,800	100

Four securities dominate the top 15: **SCON** (Feb–Mar 2000, 9 of the 15 rows), **TVIX** (Feb–Jun 2013), **CODA** (Feb 2001) and **TOPS** (Aug 2016). Volume on those days is tiny, 34 to 400 shares, against a median of 126,000.

These rows look unusual, but that alone does not prove they are errors. In this dataset **close** is already split-adjusted (KO closes at \$0.86 in 1970, section 9). When a company does many *reverse* splits, such as the leveraged ETN TVIX and the shipping company TOPS, the adjustment multiplies old prices upward and divides old volumes downward. That produces exactly this pattern: million-dollar prices with double-digit volume. SCON peaking at the height of the dot-com bubble fits the same explanation. Confirming it would need the split history of each ticker, so these rows are documented as **extreme but plausible**, not deleted.

7. Investigating adj_close

Ticker	Period	close	adj_close	Assessment
AAN	Jan–Apr 1987	\$1.06 – \$1.30	$1.5 - 1.9 \times 10^{19}$	Impossible. adj_close is $\sim 10^{19}$ times close; the adjustment factor is broken.
YRCW	1981–1986	\$72,188 – \$271,875	$\approx$ same as close	Consistent. close and adj_close agree; looks like heavy reverse-split back-adjustment.

Every one of the 20 largest adj_close values belongs to **AAN** in early 1987. The close column shows a normal price of about \$1.30 on the same days, so the two columns disagree by a factor of roughly 10^{19} . No corporate action could create that gap, which points to a broken adjustment factor in the source. **YRCW** is different: its prices are extreme, but close and adj_close agree, so the record is internally consistent.

Rows with adj_close > 100,000 or ≤ 0	By decade	Top tickers
4,059 rows above 100,000 (0.02% of data); none negative	1980s: 461 · 1990s: 574 · 2000s: 2,621 · 2010s: 403	YRCW 1,152 · ELJ 1,100 · AAN 824 · MBVX 207 · SAB 194 · TVIX 182 · CODA 155

Should they be deleted automatically? No. Following **Detect** → **Investigate** → **Document** → **Decide**, the rows were kept in the raw frame and a separate **clean_prices** frame ($0 < \text{adj_close} < 100,000$) was created for price statistics and charts. It excludes 4,060 rows: the 4,059 above 100,000 plus one row exactly at the cut-off. This keeps an audit trail and lets the decision be reversed. The threshold removes the impossible AAN and ELJ values, but it also removes YRCW, which may be genuine. That trade-off is a judgment call and is listed as a limitation.

8. Distributions and Boxplots

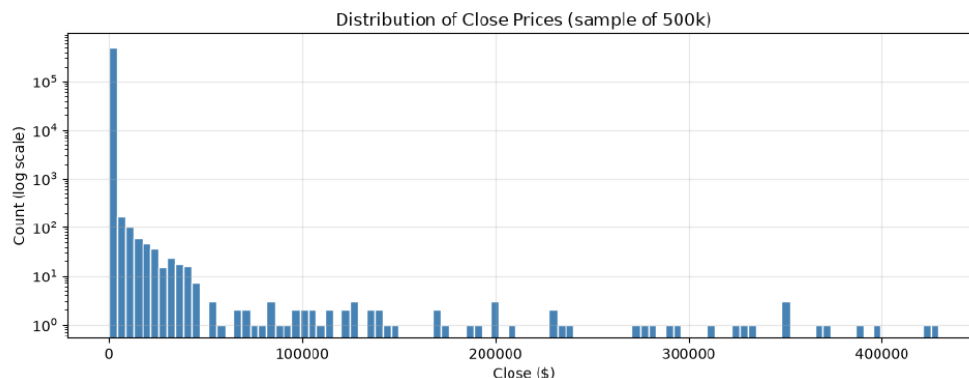


Figure 3. Histogram of close, 500,000-row random sample, log-scaled count.

The closing-price distribution is **extremely right-skewed**. Almost every observation falls in the first bin, near zero, and the tail stretches out to about \$430,000. The count axis had to be log-scaled for the tail to be visible at all. This matches **.describe()**: a 75th percentile of \$29.72 against a maximum of \$1.78 million.

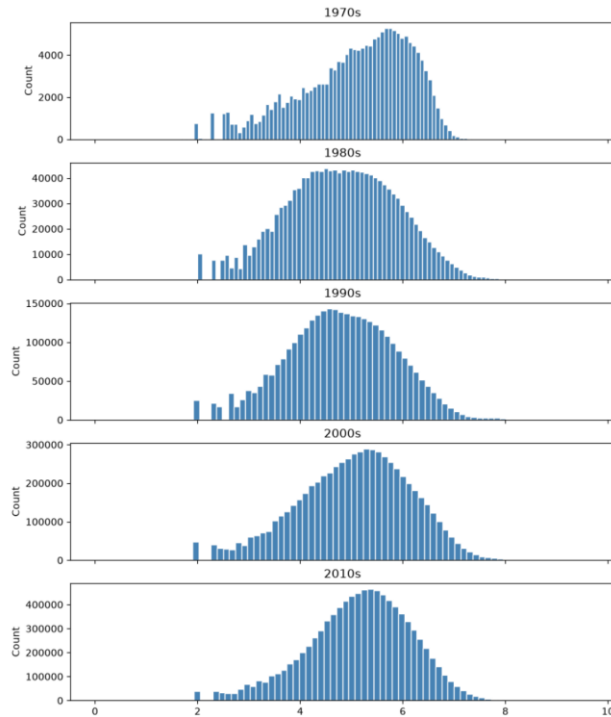


Figure 4. Histogram of $\log_{10}(\text{volume})$ for each decade.

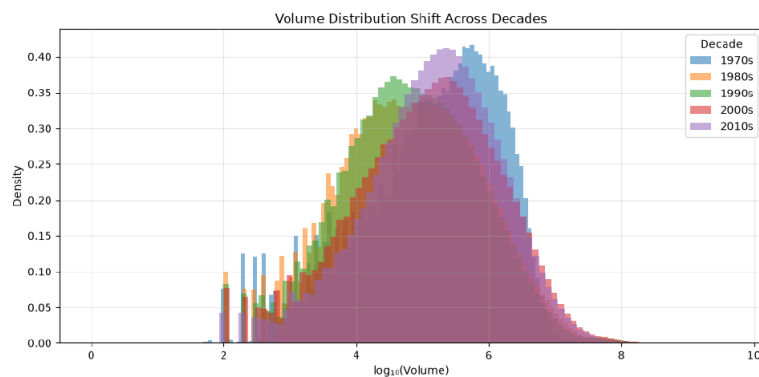


Figure 5. The same distributions overlaid (density), showing the shift between decades.

Raw volume is just as skewed, so it was plotted as $\log_{10}(\text{volume})$. On that scale the distribution becomes close to bell-shaped and centred around 5, which is 100,000 shares a day. The spike at 2 (100 shares, the standard round lot) and the empty gap just after it are artifacts of how volume is reported, not market behaviour. The overlay shows a clear shift: the 1970s peak furthest right (about $10^{5.7}$ shares) because that decade only contains 93 large, established companies; the 1980s and 1990s move left (about $10^{4.5}$) as thousands of small, thinly traded companies enter; the 2000s and 2010s move back to about $10^{5.3}$.

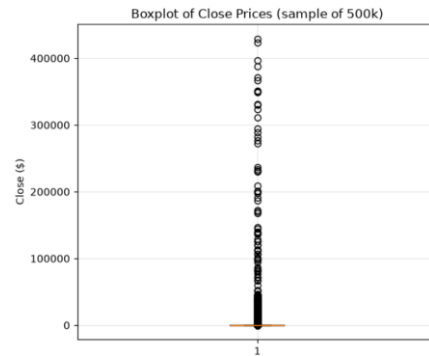


Figure 6. Boxplot of close, 500,000-row random sample.

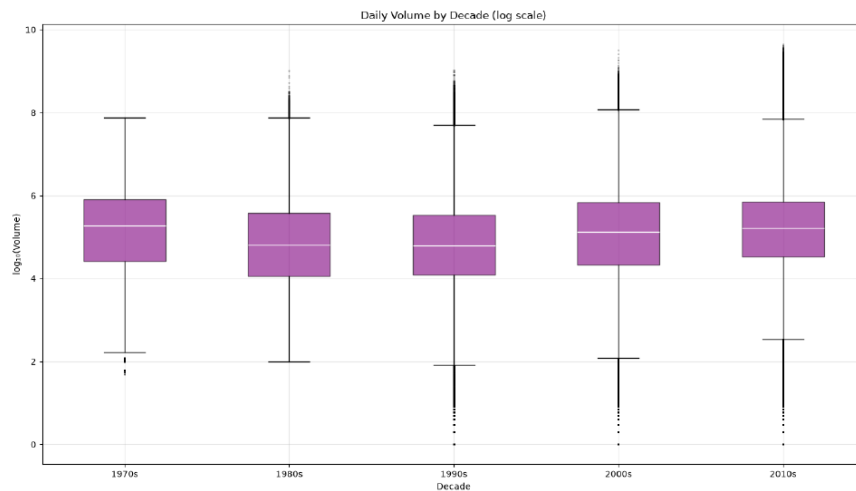


Figure 7. Boxplot of log₁₀(volume) by decade.

What the box shows: the middle 50% of values, from the 25th to the 75th percentile, with the line inside marking the median. The whiskers reach $1.5 \times$ the box height; points beyond them are flagged as potential outliers.

What the boxplots tell us: for close, the box is squashed into a flat line at zero and every visible point is an outlier, so on a raw scale the boxplot is almost unreadable. On a log scale (Figure 7) the volume boxes are stable across decades at roughly 10^4 to 10^6 shares, while the upper outliers grow each decade, reaching about $10^{9.65}$ (4.48 billion shares, the maximum in the table) in the 2010s. The lower outliers near 0 are days with only 1 to 10 shares traded.

Boxplot vs. histogram: the histogram shows the shape of the whole distribution; the boxplot summarises it in five numbers and makes outliers easy to count and compare between groups. Financial data is full of extreme values, so a quick visual of where "normal" ends is especially useful.

9. One Stock Over Time: KO



Figure 8. Coca-Cola (KO) daily closing price, 1970–2018.

KO (Coca-Cola) was selected because it trades across the full 1970–2018 period. The close starts below \$1 because it is split-adjusted, stays flat through the 1970s and early 1980s, then climbs steadily from about 1985 to a peak near **\$44 in 1998**. It falls back to around \$20 during 2000–2005, drops again in 2008–09, and then recovers to a new high of about \$47 by 2018.

The unusually high period is 1997–98; the lows after it are 2003–05 and early 2009. There are no long gaps: the line is continuous, and the only missing days are weekends and market holidays, when no trading happens. To explain the big moves we would need information that is not in these files: earnings reports, dividends, split dates, company news, and broad market events such as the 2008 financial crisis.

10. Comparison by Decade

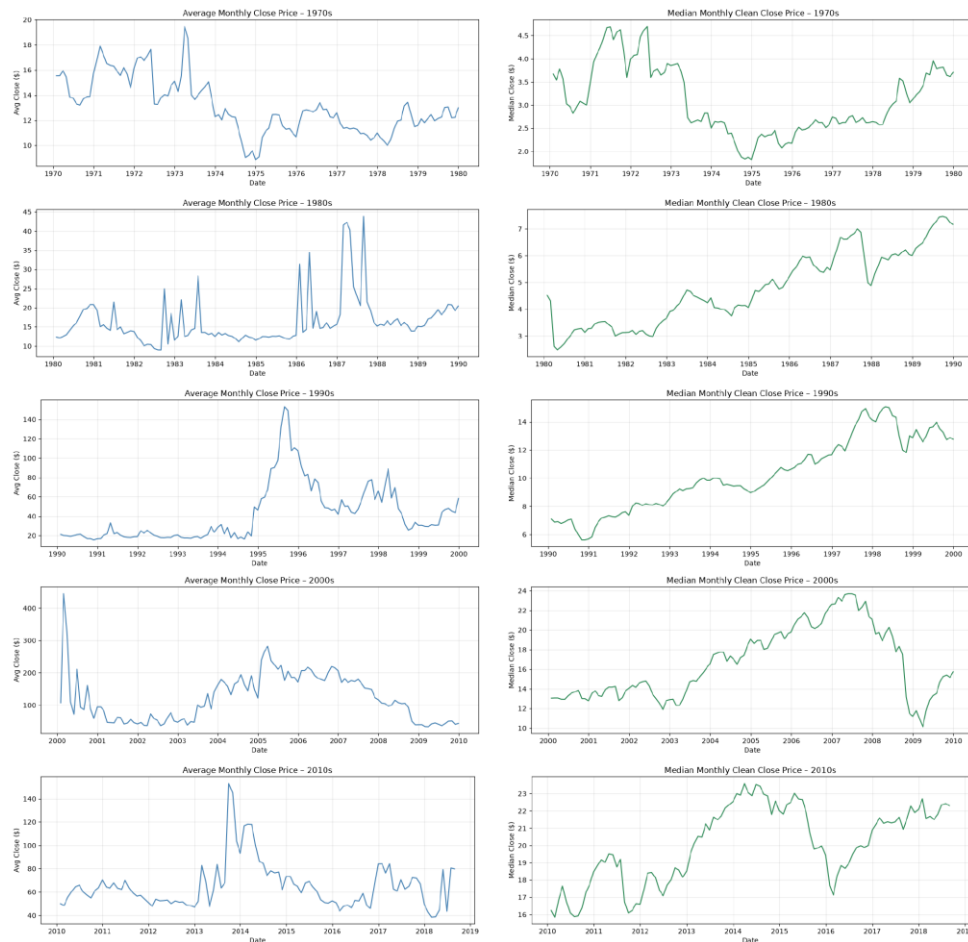


Figure 9. Left: average monthly close (raw data). Right: median monthly close (clean_prices).

The mean-based chart (left) is full of one-month spikes, for example February 2000 at about \$440 and several jumps in 1986–87. Thousands of stocks cannot all double and half in alternate months, so these spikes come from a few extreme tickers entering and leaving the average. The median of the cleaned data (right) removes them and shows four real market declines:

Event	What the median chart shows (approx.)
1973–74 bear market	From about \$4.70 in 1972 to about \$1.85 by late 1974, a fall of roughly 60%.
1987 crash	Peak near \$7.00 in Aug 1987, down to about \$4.90 by year end. Invisible in the mean chart.
2000–02 dot-com bust	Only a mild dip for the typical stock; the bubble was concentrated in a few tech names.
2007–09 financial crisis	From about \$23.70 in 2007 to about \$10.20 in early 2009, a fall of about 57%.

Decade	Rows	Tickers	Median adj_close	Mean adj_close	CV	Median volume	Mean/median volume
1970s	164,499	93	0.35	7.31	5.46	187,200	3.87
1980s	1,470,110	900	1.18	9.22	19.85	64,700	12.95
1990s	3,427,258	2,117	4.87	30.29	25.72	62,200	13.89
2000s	6,569,431	3,429	11.65	58.05	18.22	133,500	10.37

Decade	Rows	Tickers	Median adj_close	Mean adj_close	CV	Median volume	Mean/median volume
2010s	9,338,531	5,685	17.79	52.54	13.32	165,400	7.98

Summary by decade from `clean_prices`. CV = standard deviation ÷ mean.

The median adjusted close rises every decade, from \$0.35 to \$17.79. Early values are tiny because `adj_close` expresses old prices in today's share terms after splits and dividends. The coefficient of variation peaks in the 1990s (25.7) and then falls to 13.3, so prices became less spread out relative to their average. Volume moves the other way: the median trading day is actually lower in the 2010s than in the 1970s (165,400 vs 187,200 shares), but the gap between the most and least traded stocks widened, with volume standard deviation rising from 1.4 million to 17.7 million. Finally, exactly **93 tickers appear in all five decades**, the same as the number of tickers in the 1970s. Every 1970s company in the file survived to 2018, which is direct evidence of survivorship bias.

11. Summary of Findings

Data structure. Two datasets linked one-to-many by ticker: a 6,460-row reference list with exchange, sector and industry, and a 20,973,889-row daily price file covering 1970-01-02 to 2018-08-24 on NASDAQ and NYSE.

Data quality. No missing values or duplicates in the price file. 1,440 securities lack sector and industry and were kept. One accidental header row (ticker = "Symbol", sector = "SECTOR") was found in the stocks file and removed, leaving 6,459 securities in 12 sectors. The date column was converted to datetime.

Numerical analysis. All price columns and volume are strongly right-skewed. Mean close (\$76.11) is almost five times the median (\$15.45), so medians are used for the typical stock.

Outliers and anomalies. Extreme close values (SCON, TVIX, CODA, TOPS) are consistent with reverse-split adjustment and were kept. Extreme `adj_close` values were investigated ticker by ticker: AAN (and ELJ) are impossible, YRCW is consistent but extreme. 4,060 rows were excluded from price statistics in a separate `clean` frame; the raw data was not changed.

Visual analysis. Histograms and boxplots confirm the heavy right skew and show volume is roughly log-normal around 100,000 shares. KO shows long-term growth with clear 1998 and 2008 turning points. Median charts reveal four real market declines that the mean hid.

Cleaning decisions

Issue	Rows affected	Decision	Reason
Missing sector / industry	1,440 securities	Keep	Valid securities; missing ≠ wrong
Header row (ticker "Symbol")	1	Removed	Describes the file, not a security
Duplicates (full / ticker+date)	0 / 0	None needed	Checked both levels
Extreme close (SCON, TVIX...)	Top values	Keep	Plausible reverse-split adjustment
<code>adj_close</code> ≥ 100,000 or ≤ 0	4,060	Exclude in <code>clean_prices</code> only	AAN/ELJ impossible; raw data kept

Limitations

Survivorship bias: only companies still listed in 2018 appear, so failed firms are missing and long-run results look better than reality. **Changing composition:** each decade contains a different set of tickers, so

decade differences mix real market change with sampling change. **Nominal dollars:** no inflation adjustment. **Threshold choice:** the 100,000 cut-off also removes YRCW, which may be genuine.

Questions for further analysis

Does median volatility rise sharply around the four declines found in section 10? Do sectors react differently to the same crisis, for example Finance in 2008 versus Technology in 2000? How much of the decade-to-decade price growth remains if the analysis is restricted to the 93 companies present in every decade? These questions lead naturally into the feature engineering and modelling in Projects 2 and 3.